IPCON® PPA

Polyphthalamide Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.05 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	365 m ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C) or PC (max heat resistance: 130 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm
Recommendation for body protection during use	No special requirements

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.14 g/cm³
Shrinkage Rate*	Specially-defined	0.16% - 0.43%
Surface Hardness	Shore hardness	87D
Saturated Water Absorption Rate	25 °C, 55% RH, room air	1.32%
Melt Index	290 °C, 2.16 kg	15.6 g/10 min
Melting Temperature	DSC, 10 °C/min	246 °C
Vicar Softening Temperature	ISO 306, GB/T 1633	Not annealed: 106 °C; annealed: 133 °C
Heat Deflection Temperature	ISO 75, 0.45 MPa	Not annealed: 103 °C; annealed: 131 °C
Relative Thermal Index (RTI) Max Continuous Service Temperature	Actual working conditions simulation test	100 - 125 °C (dry state) (Related to load, size, shape, wall thickness, rate of water content, etc., and the value after annealing is higher than that before annealing in general)

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527, GB/T 1040	76 MPa
Tensile Strength Z	ISO 527, GB/T 1040	40 MPa
Young's Modulus XY	ISO 527, GB/T 1040	2863 MPa
Young's Modulus Z	ISO 527, GB/T 1040	2677 MPa
Breaking Elongation Rate XY	ISO 527, GB/T 1040	3.4%
Breaking Elongation Rate Z	ISO 527, GB/T 1040	1.6%
Flexural Strength XY	ISO 178, GB/T 9341	123 MPa
Flexural Strength Z	ISO 178, GB/T 9341	62 MPa

Flexural Modulus XY	ISO 178, GB/T 9341	2672 MPa
Flexural Modulus Z	ISO 178, GB/T 9341	2576 MPa
Impact Strength XY	ISO 179, GB/T 1043	32.8 kJ/m²
Impact Strength Z	ISO 179, GB/T 1043	5.4 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 290 °C, bed temperature: 100 °C, printing speed: XY—150 mm/s, Z—100 mm/s) and IPCON PPA (black) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The XY specimens' annealing parameter is: 100 °C, 4 h → 120 °C, 8 h. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Long-term heat resistance of 120 °C (after annealing); smooth surface; high precision, strength and toughness; moisture and creep-resistant; ultra-low warping; compatible with AMS or CFS; 4 basic colors: black, white, gray, and red.

2. Can be used to print bicycle accessories, temperature-resistant and humidity-resistant insulating parts for automobiles, electronics and mechanical equipment, as well as high wear-resistant parts such as gears, slide rails, and shaft sleeves.

• Recommended Printing Settings

Drying Settings Before Printing	100 - 130 °C, 10 - 14 h (Blast drying oven)
Printing Temperature and Humidity	≤ 100 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C, ≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PA
Compatible Printer Type	Enclosed-frame (recommended), open-frame,
AMS or CFS	Compatible
Compatible Nozzle Material	Any common material, such as brass, stainless steel, hardened steel, diamond, tungsten carbide, etc
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Apply liquid glue or PVP solid glue stick if the first layer bonding is not firm enough
Nozzle Temperature	280 - 310 °C
Bed Temperature	100 - 120 °C
Fan Speed	0% - 40%
Max Print Speed	300 mm/s

IPCON Polymer Material (Suzhou) Co., LTD

Address: Room 632, Building E, Dongchuang Science Park, No. 216, Jinfeng Road, Mudu Town,Wuzhong District, Suzhou City,China

Web Site: www.ipconpolymer.com

E-mail: sales@ipconpolymer.com

Tel: 400-139-1152

Innovative pragmatic Consistent